



Module Presenter's Manual

for

Web Component Development
using Jakarta EE

Effective from: June, 2024
Ver. 1.0

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	June 2024	New	-

Table of Contents

Sr. No.	Details	Page No.
1.	Introduction	1
2.	Information on Session Allocation	2
3.	Module Deliverables on OnlineVarsity	2
4.	Week-wise Session Schedule	3
5.	Session Coverage	4
6.	Library References	11

1. Introduction

At the end of this course, students will be able to:

- Understand the basics of Web application development
- Understand the Jakarta EE 10 platform
- Explain Servlets and their lifecycle
- Understand and create robust Web applications using MVC architecture, session management, filters, and database integration
- Explain Jakarta Server Pages
- Learn to develop JSP components
- Understand and create easy to maintain view components using JSP pages, the Expression Language, and elements of the JSP Standard Tag Library (JSTL)
- Create secure Web applications using the features of the Jakarta EE Web container

2. Information on Session Allocation

Module	Online Hours
Web Component Development using Jakarta EE	32

Throughout this Presenter's Manual, the module **Web Component Development using Jakarta EE** will be referred to as **JSP**.

3. Module Deliverables on OnlineVarsity

To aid the teaching process, following are the deliverables:

Resources available on Onlinevarsity:

Feature - Description/Functionality
Ebook -Student reading material in the form of PDF.
Download eBook - Student has the option to download the subject related e-book and read offline.
Practice 4 Me - Student can test and evaluate their understanding of module related topics.
Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
References - Student can access additional subject related material for reading.

4. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 JSP – TL1	Session 2 JSP – TL2	Session 3 JSP – TL3	Session 4 JSP – TL4
2	Session 5 JSP – TL5	Session 6 JSP – TL6	Session 7 JSP – TL7	Session 8 JSP – TL8
3	Session 9 JSP – TL9	Session 10 JSP – TL10	Session 11 JSP – TL11	Session 12 JSP – TL12
4	Session 13 JSP – TL13	Session 14 JSP – TL14	Session 15 JSP – TL15	Session 16 JSP – TL16

JSP: Web Component Development using Jakarta EE

TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	JSP – TL1	All the topics as listed below from Session 1 and 2 of Jakarta Server Pages and Servlets book should be covered in this session. <u>Session 1 - Introduction to Web Applications</u> ➤ Explain Web applications and their advantages ➤ Describe the architecture and components of a Web application ➤ Describe the role of HTTP protocol and its methods used for accessing Web pages ➤ Describe the HTTP/2 protocol and its methods used for accessing Web pages ➤ Explain different types of components used in developing a Web application ➤ List the advantages and disadvantages of Servlets ➤ Describe the sequence of events from request to response in a Web application ➤ Explain the services provided by a Web container <u>Session 2 – Introduction to Jakarta EE</u> ➤ Explain the Jakarta EE platform ➤ Describe Jakarta EE architecture and profiles ➤ Describe Jakarta EE application model and its components ➤ Identify the microservices and cloud-native principles of Jakarta EE ➤ List advantages and disadvantages of Jakarta EE ➤ Explain the key Jakarta EE specifications ➤ Explain the installation of Jakarta EE components ➤ Explain the installation of JDK and the process of setting up the system variables ➤ Identify the features and capabilities of Eclipse IDE ➤ Explain the installation of Eclipse IDE	Jakarta Server Pages and Servlets SG-Session 1 and 2 XP-Session 1 and 2 TG-Session 1 and 2
2	JSP – TL2	All the topics as listed below from Session 3 of Jakarta Server Pages and Servlets book should be covered in this session. <u>Session 3 – Developing and Deploying a Web Application</u>	Jakarta Server Pages and Servlets SG-Session 3 XP-Session 3 TG-Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain Apache Tomcat server ➤ Describe the lifecycle and directory structure of a Web application ➤ Describe the process of Web application development with Eclipse IDE ➤ Describe directory structure in Eclipse ➤ Explain how to package a Web application ➤ Describe the process to build and deploy a simple Web application using Eclipse and Apache Tomcat server	
3	JSP – TL3	All the topics as listed below from Session 4 of Jakarta Server Pages and Servlets book should be covered in this session. Session 4 – Jakarta Servlets ➤ Explain the Jakarta Servlet API ➤ Explain the servlet architecture and life cycle of Servlet ➤ Describe the servlet mapping specifications ➤ Explain methods of ServletRequest and HttpServletRequest interfaces ➤ Explain methods of ServletResponse and HttpServletResponse interfaces ➤ Describe use of response headers ➤ Explain how to read text and binary data from a request ➤ Explain the ServletConfig and ServletContext interface ➤ Explain Servlet context Attribute ➤ Explain redirection of client requests ➤ Explain RequestDispatcher interface ➤ Explain error handling in Servlet ➤ Explain uploading files with Jakarta Servlet Technology	Jakarta Server Pages and Servlets SG-Session 4 XP-Session 4 TG-Session 4
4	JSP – TL4	The Try It Yourself questions of Session 1 to Session 4 of Jakarta Server Pages and Servlets book should be covered in this session.	Jakarta Server Pages and Servlets Session 1-4
5	JSP – TL5	All the topics as listed below from Session 5 and Session 6 of Jakarta Server Pages and Servlets book should be covered in this session. Session 5 – Session Tracking ➤ Describe the stateless nature of HTTP and HTTP/2 protocols	Jakarta Server Pages and Servlets SG-Session 5 & 6 XP-Session 5 & 6 TG-Session 5 & 6

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain the necessity of tracking client identity and state ➤ Explain the URL rewriting method for session tracking ➤ Explain how to use hidden form fields ➤ Explain the use of Cookie class and its methods ➤ Explain how to store and retrieve information in a session ➤ Describe the use of HttpSession interface and its methods ➤ Explain how to invalidate a session ➤ Explain invalidating a session with Eclipse ➤ Describe Java API with JSON binding ➤ Explain JSON serialization <u>Session 6 – Jakarta Server Pages</u> ➤ Define Jakarta Server Pages ➤ Outline the differences between servlet and JSP ➤ Contrast Jakarta Server Pages against Java Server Pages ➤ Explain the necessity for JSP ➤ List the benefits of JSP ➤ Identify the situations where to use JSP and servlets ➤ Explain JSP architecture ➤ Identify various phases in the life cycle of a JSP page ➤ Explain various elements in JSP ➤ List and explain the use of various directives in JSP ➤ Describe changes in JSP 3.1 ➤ Explain procedure for Integrating Jakarta servlet with JSP ➤ Describe dispatching request from servlet to JSP	
6	JSP – TL6	All the topics as listed below from Session 7 of <i>Jakarta Server Pages and Servlets</i> book should be covered in this session. <u>Session 7 – Filters and Annotations</u> ➤ Explain the concept and working of filters ➤ Describe the Servlet filter configuration ➤ List the benefits of filters ➤ Explain the Filter API interfaces and their methods ➤ Explain Filter to servlet mapping	Jakarta Server Pages and Servlets SG-Session 7 XP-Session 7 TG-Session 7

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain the use of wrapper classes in managing Servlets ➤ Explain how to alter a request and response using filters ➤ Describe the basic concept of annotations ➤ List different types of annotations supported in Servlet API ➤ Outline the steps to upload the file using HTML form elements ➤ Explain the mechanism to upload files on the server using Servlet ➤ Describe Server push in Servlet 6.0	
7	JSP –TL7	All the topics as listed below from Session 8 of Jakarta Server Pages and Servlets book should be covered in this session. <u>Session 8 – Database Access and Event Handling</u> ➤ Explain database handling using JDBC 4.3 ➤ Describe JPA and its role ➤ Describe connecting database using JPA ➤ Explain the significance of session handling and session events ➤ Explain different types of listener interfaces used in Servlets ➤ Explain JDBC Vs JPA	Jakarta Server Pages and Servlets SG-Session 8 XP-Session 8 TG-Session 8
8	JSP – TL8	All the topics as listed below from Session 9 of Jakarta Server Pages and Servlets book should be covered in this session. <u>Session 9 – Asynchronous Servlet Communication</u> ➤ Explain necessity for Asynchronous Servlet ➤ Describe advantages and use of Asynchronous servlet ➤ Explain how to create Asynchronous Servlet and Asynchronous Listener ➤ Explain how to create an asynchronous JavaScript client using XMLHttpRequest object ➤ Explain the necessity of non-blocking I/O support in Servlet ➤ Explain how to implement non-blocking I/O in asynchronous Servlet ➤ Explain necessity for protocol upgrade ➤ Elaborate on the process of protocol upgrade in a Servlet	Jakarta Server Pages and Servlets SG-Session 9 XP-Session 9 TG-Session 9

Session No.	Session Title	Session Details	Deliverables' Mapping
9	JSP – TL9	The Try It Yourself questions of Session 5 to Session 9 of Jakarta Server Pages and Servlets book should be covered in this session.	Jakarta Server Pages and Servlets Session 5-9
10	JSP – TL10	All the topics as listed below from Session 10 of Jakarta Server Pages and Servlets book should be covered in this session. <u>Session 10 – JSP Implicit Objects and Standard Actions</u> ➤ Explain the concept of implicit objects in JSP ➤ List various types of implicit objects in JSP ➤ Explain how to use the request and response objects ➤ Identify how to use the out object ➤ Describe how to use the session object ➤ Explain the application object ➤ Identify the use of pageContext object ➤ Explain how to use the page and config objects ➤ Outline the concept of standard actions in JSP ➤ Describe how to use the <jsp:include> and <jsp:forward> elements	Jakarta Server Pages and Servlets SG-Session 10 XP-Session 10 TG-Session 10
11	JSP – TL11	All the topics as listed below from Session 11 of Jakarta Server Pages and Servlets book should be covered in this session. <u>Session 11 – Model-View-Controller Architecture</u> ➤ Describe JSP models ➤ Describe the advantages and disadvantages of Model1 ➤ Describe the advantages and disadvantages of Model2 ➤ Explain the Model-View-Controller (MVC) architecture ➤ Explain the relationship between the components of MVC ➤ Explain Controller and its use ➤ Explain View and its use ➤ Explain Model and its use ➤ Identify how to develop a Web application based on MVC architecture ➤ Describe MVC and CDI ➤ Identify MVC events	Jakarta Server Pages and Servlets SG-Session 11 XP-Session 11 TG-Session 11

Session No.	Session Title	Session Details	Deliverables' Mapping
12	JSP – TL12	All the topics as listed below from Session 12 of Jakarta Server Pages and Servlets book should be covered in this session. Session 12 – JSP Expression Language ➤ Explain how to use script expressions in JSP ➤ Describe the implicit objects used in EL ➤ Describe various operators used in EL ➤ Explain how to create static method and tag library descriptor using EL ➤ Explain how to access EL functions within JSP ➤ Explain the concept of boxing and unboxing ➤ Explain how to coerce a value to string or number type	Jakarta Server Pages and Servlets SG-Session 12 XP-Session 12 TG-Session 12
13	JSP – TL13	The Try It Yourself questions of Session 10 to Session 12 of Jakarta Server Pages and Servlets book should be covered in this session.	Jakarta Server Pages and Servlets Session 10-12
14	14	Session 13 - Standard Tag Library and Internationalization ➤ List the advantages of using JSTL ➤ Describe different tag libraries available in JSTL ➤ Explain how to configure JSTL library in Eclipse ➤ Explain general-purpose tags ➤ Explain decision-making in the tags ➤ Explain iteration tags in the core tag library ➤ List different tags available in the SQL tag library ➤ Recall in brief concept of internationalization and localization ➤ Explain the resource bundling mechanism and resource bundle for various locales ➤ Explain various formatting techniques in servlets for internationalization ➤ Explain Reactive JAX-RS Client API	Jakarta Server Pages and Servlets SG-Session 13 XP-Session 13 TG-Session 13
15	JSP – TL15	All the topics as listed below from Session 14 and Session 15 of Jakarta Server Pages and Servlets book should be covered in this session. Session 14 – Securing Java Web Applications ➤ Explain the necessity for and features of securing Web applications ➤ Describe Jakarta EE Security API	Jakarta Server Pages and Servlets SG-Session 14 & 15 XP-Session 14 & 15 TG-Session 14 & 15

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Describe the HTTP basic, digest, client, and form-based authentication method of ensuring security ➤ Explain how to configure users in Tomcat ➤ Explain how to specify authentication mechanisms using annotations ➤ Describe the steps to implement declarative security ➤ Explain the steps to implement programmatic security ➤ Describe the HttpServletRequest methods for identifying users ➤ Explain the use of SSL certification <u>Session 15 – Building an Application Using JSP and Servlet</u> ➤ Explain the process of creating a Dynamic Web application using JSP and Servlet ➤ Describe connecting a database using JPA ➤ Describe the steps to connect the JSP application to a MySQL database	
16	JSP – TL16	The Try It Yourself questions of Session 13 to Session 15 and Appendix of Jakarta Server Pages and Servlets book should be covered in this session.	Jakarta Server Pages and Servlets Session 13-15 Appendix

6. Library References

- | |
|---|
| ➤ Murach's Java Servlets and JSP, 3rd Edition by by Andrea Steelman
Michael Urban |
| ➤ Java EE to Jakarta EE 10 Recipes: A Problem-Solution Approach for Enterprise Java by Josh Juneau and Tarun Telan |

~~ End of Document ~~